AS-Regularized Deep Market Making without Hybrid Reward Shaping

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Abstract

Deep reinforcement-learning market makers are often trained with hybrid rewards that mix dampened profit-and-loss, transaction advantage, and inventory penalties. We revisit the Attn-LOB / C-PPO market maker of Guo et al. [6] and ask whether the behavioural prior can instead come from a classical quoting model. We propose SOFT-AS PPO: PPO with pure profit as the economic reward and a soft action-space regularizer toward a per-stock Avellaneda–Stoikov (AS) teacher fitted on training data. On a synthetic, paper-shaped replication benchmark, the selected soft-AS policy improves mean episode PnL over reward-shaped C-PPO by 15.10 across 15 stock–seed pairs (95% CI $[9.19, 21.01]$, $p = 8.1 \times 10^{-5}$) and over PROFIT PPO by 9.50 (95% CI $[5.45, 13.56]$, $p = 1.9 \times 10^{-4}$). It also stays much closer to the AS teacher in action space. The tradeoff is visible: SOFT-AS PPO quotes tighter, fills more, and carries more inventory, especially under high latency. In this benchmark, AS regularization gives a cleaner and more inspectable way to steer a deep market maker than hybrid reward shaping.

1 Introduction

Market making is the problem of continuously quoting bid and ask limit orders to earn the spread while managing inventory risk [15, 11]. A modern limit order book (LOB) is high-dimensional, noisy, and fast-moving, so recent work uses deep reinforcement learning (RL) to learn quoting policies directly from LOB state. The Attn-LOB / C-PPO agent of Guo et al. [6] is a strong example: a convolutional-attention encoder reads a rolling LOB window, and PPO emits continuous bid/ask quote parameters.

The agent's behaviour, however, is largely determined by a hand-designed reward. The original C-PPO reward combines dampened PnL, trading PnL, and an inventory penalty:

$$R_t = \underbrace{\Delta\mathrm{PnL}_t - \max(0, \eta\Delta\mathrm{PnL}_t)}_{\text{dampened PnL}} + \underbrace{X_{t,v}(p_{m,t} - X_{t,p})}_{\text{trading PnL}} - \underbrace{\zeta\mathrm{Inv}_t^2}_{\text{inventory penalty}} . \tag{1}$$

These terms are useful engineering, but they are not potential-based reward shaping [9]; they change the optimization target and introduce coefficients that must be retuned for each market.

[*]Corresponding author. Code and configuration files are available at `https://github.com/pierre-roth/mlfcs-gapa`.

Our alternative keeps the accounting simpler. The environment reward is pure profit, while the behavioural prior comes from Avellaneda–Stoikov (AS), a classical market-making model that gives an interpretable reservation price and spread from inventory, volatility, time remaining, risk aversion, and liquidity [1]. Instead of copying AS, we regularize the learned continuous action toward a fitted AS teacher:

$$J(\pi) = \mathbb{E}_\pi\left[\sum_t \Delta\mathrm{PnL}_t - \lambda \left\| a_t^\pi - a_t^{\mathrm{AS}} \right\|_2^2\right]. \tag{2}$$

The AS term is part of the training objective, but the economic reward being optimized is ordinary marked-to-market profit.

Contributions. We formulate AS guidance as a soft action-space regularizer for continuous-action PPO market making, calibrate one AS teacher per stock from the training window, and keep the extension code separated from the paper-replication pipeline. The experiments use a matched synthetic benchmark following the original paper's stock/day/episode/action structure, with five PPO seeds for SOFT-AS PPO, C-PPO, and PROFIT PPO on three synthetic stocks. We also regenerate the original-style analyses–performance by stock, latency, attention, and decision traces–for the proposed method, and add teacher-action divergence diagnostics.

2 Related work

Classical market-making models derive quotes by balancing spread capture and inventory risk. In the AS model, with inventory q, mid-price s, volatility σ, risk aversion γ, liquidity κ, and horizon T, the reservation price and total spread are

$$r(s,q,t) = s - q\gamma\sigma^2(T-t), \qquad \delta^a + \delta^b = \gamma\sigma^2(T-t) + \frac{2}{\gamma}\log\Big(1+\frac{\gamma}{\kappa}\Big). \tag{3}$$

This structure is interpretable but depends on assumptions about price and arrival processes [4]. Deep LOB models such as DeepLOB and related architectures learn representations directly from order-book tensors [17, 16, 10]. RL market makers have used handcrafted state, discrete actions, and model-free policies [8, 14, 18, 12, 3]. Guo et al. [6] combine an Attn-LOB encoder with a continuous, AS-like action space and the hybrid reward in Eq. (1). We keep their encoder, simulator, action space, and PPO training setup, but move the behavioural structure from the hybrid reward into an AS action prior.

3 Method

Replicated agent. The state is a rolling 50×40 tensor of the top-10 ask/bid prices and volumes, plus agent state such as inventory and remaining episode time. The Attn-LOB feature extractor applies convolutional filters, an Inception block, and multi-head self-attention. The continuous policy emits two normalized actions, $A_1, A_2 \in [0,1]$: a reservation-price bias and a quoted spread,

$$\delta = A_1 \cdot \text{max\_bias}, \quad p_r = p_m - \mathrm{sign}(\mathrm{Inv})\delta, \quad \text{spread} = A_2 \cdot \text{max\_spread}, \quad p_{a,b} = p_r \pm \frac{\text{spread}}{2}. \tag{4}$$

We retain the paper's max_bias $= 0.05$, max_spread $= 0.1$, minimum_trade_unit $= 100$, and end-of-episode liquidation.

AS teacher. At each step we convert Eq. (3) into the same normalized action coordinates as Eq. (4):

$$a_t^{\mathrm{AS}} = \left(\mathrm{clip}\left(\frac{|q|\gamma\hat{\sigma}^2(T-t)}{\text{max\_bias}}, 0, 1\right), \mathrm{clip}\left(\frac{\delta^a+\delta^b}{\text{max\_spread}}, 0, 1\right)\right). \tag{5}$$

The teacher is deterministic once (γ, κ) are fitted. We estimate one teacher per stock from the training data: κ from the mean level-1 half-spread and γ from realized volatility, then use a lower-risk $\gamma = 0.025$ for the selected policy. The resulting κ values are 125.2, 23.9, and 83.8 for stocks 000001, 000858, and 002415.

Table 1: Five-seed matched comparison. Values average 15 stock–seed units per method. PnL and Sharpe include standard errors across those units.

Method	PnL ($\pm$SE)	ND-PnL	Sharpe ($\pm$SE)	Abs. inv.	Spread	Fills
SOFT-AS PPO	$\mathbf{75.17} \pm 22.06$	$\mathbf{1223.9}$	$\mathbf{2.54} \pm 0.44$	54.2	0.0468	511.7
PROFIT PPO	65.67 ± 22.51	1010.5	2.29 ± 0.42	26.6	0.0551	437.7
C-PPO	60.07 ± 21.38	812.1	2.07 ± 0.51	19.3	0.0637	333.1

Methods compared. The main paper reports three PPO methods. C-PPO is the reward-shaped baseline of Eq. (1). PROFIT PPO removes the hybrid reward and trains on pure profit only. SOFT-AS PPO trains on the profit objective plus the AS action regularizer in Eq. (2), with $\lambda = 0.10$ and the low-risk per-stock teacher. Hard AS constraints, BC warm start, direct AS, and λ sensitivity are appendix-only controls.

4 Experimental setup

A synthetic benchmark. The original Shenzhen Stock Exchange LOB data are proprietary. We therefore use a synthetic event-level LOB generator that preserves the experimental shape of Guo et al. [6]: three stock identities, top-10 book levels, stable intraday windows, 10 train days, 11 test days, and 2000-event episodes. Each instrument has its own base price and generated training/test days. The latent fair value moves through stable, trending, and volatile regimes; around it the generator builds a top-10 book, trade stream, and market-order flow. The simulator then replays this stream and fills the agent's quotes only when recorded trades cross the quote. There is no market impact and no transaction cost. The benchmark is useful for controlled comparisons and debugging because every method faces the same synthetic tapes, but it remains a proxy for the proprietary data used in the original paper.

The original code trains one model per stock, not a single shared cross-stock policy, and we keep that design. The main comparison uses five PPO seeds per stock for each of C-PPO, PROFIT PPO, and SOFT-AS PPO: 15 paired stock–seed units per method. All learned methods use the same state, encoder, action space, trade granularity, train/test split, and 400k PPO training steps.

For an episode with final marked-to-market value V_T and initial value V_0, $\mathrm{PnL} = V_T - V_0$. We define $\mathrm{ND\text{-}PnL} = \mathrm{PnL}/(\overline{s} + 10^{-7})$, where $\overline{s}$ is the mean quoted spread in the episode. Sharpe is computed on event-to-event marked-value increments ΔV_t as $\sqrt{n}\,\overline{\Delta V}/\mathrm{sd}(\Delta V)$, with no annualization. Paired tests use stock–seed units, not individual episodes.

5 Results

5.1 Soft AS improves return over both controls

Table 1 and Figure 1 give the main result. SOFT-AS PPO is best on mean PnL, ND-PnL, and Sharpe. The profit-only control is also stronger than the original reward-shaped baseline, so part of the improvement comes from removing the old hybrid reward. The AS regularizer adds a second gain on top of that stronger profit-only baseline.

The paired tests in Table 2 are the cleanest comparison because each row compares methods on the same stock and seed. SOFT-AS PPO improves PnL over C-PPO by 15.10 with a 95% CI of $[9.19, 21.01]$ and $p = 8.1 \times 10^{-5}$. Against PROFIT PPO, the improvement is 9.50 with $p = 1.9 \times 10^{-4}$. This matters because PROFIT PPO already removes the hybrid reward; the fitted AS prior still adds statistically visible signal.

Figure 2 shows why paired tests matter. The high-price stock 000858 dominates raw PnL, but the proposed method is not a single-stock artifact. It improves strongly on 000001 and 002415, and remains close to the best result on 000858.

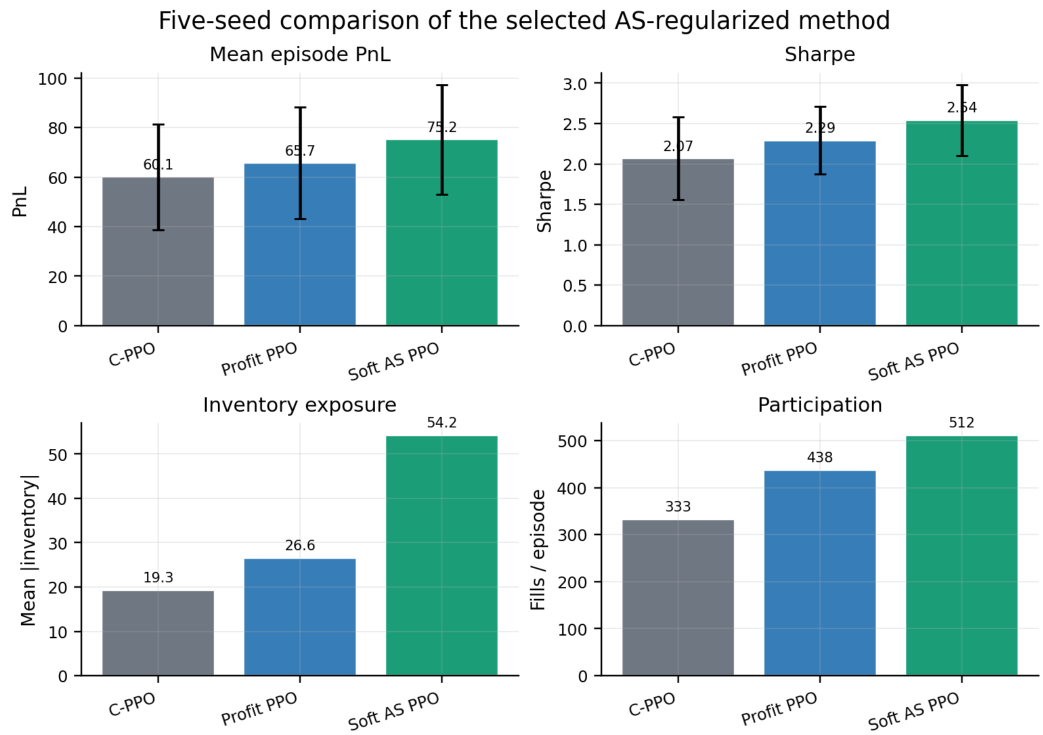

Figure 1: Headline performance and risk diagnostics over five seeds. SOFT-AS PPO has the strongest PnL and Sharpe, but reaches them by quoting tighter and participating more, which raises inventory exposure.

Table 2: Paired PnL comparisons over 15 stock–seed units.

Method	Baseline	ΔPnL	95% CI	p	Win rate
SOFT-AS PPO	C-PPO	15.10	$[9.19, 21.01]$	8.1×10^{-5}	0.93
PROFIT PPO	C-PPO	5.60	$[0.36, 10.83]$	0.038	0.73
SOFT-AS PPO	PROFIT PPO	9.50	$[5.45, 13.56]$	1.9×10^{-4}	0.93

5.2 The AS prior changes behaviour directly

The main advantage of AS regularization over reward shaping is inspectability: at evaluation time we can log both the policy action and the AS teacher action. Figure 3 shows that SOFT-AS PPO is substantially closer to the teacher in both quote-bias and spread coordinates. The mean L2 action gap falls from 0.468 for C-PPO and 0.469 for PROFIT PPO to 0.230 for SOFT-AS PPO.

The same diagnostic run shows more fills per episode for SOFT-AS PPO (96.8 versus 54.4 for C-PPO at the logged evaluation granularity) and larger inventory excursions. In the five-seed training evaluation, SOFT-AS PPO also has higher average inventory and more fills than both controls (Table 1). AS regularization improves the policy by making it more willing to quote, so the added return comes with a clearer inventory and latency tradeoff.

5.3 Latency remains the main weakness

The original paper studied latency, so we reran an evaluation-only latency sweep for the saved seed-0/1/2 checkpoints at 1, 5, 10, 20, and 50 events. Figure 4 shows the key caveat. At low latency, SOFT-AS PPO has the best return. As latency grows, all methods degrade, and the more active SOFT-AS PPO policy degrades faster because stale quotes and larger inventory become costly. This

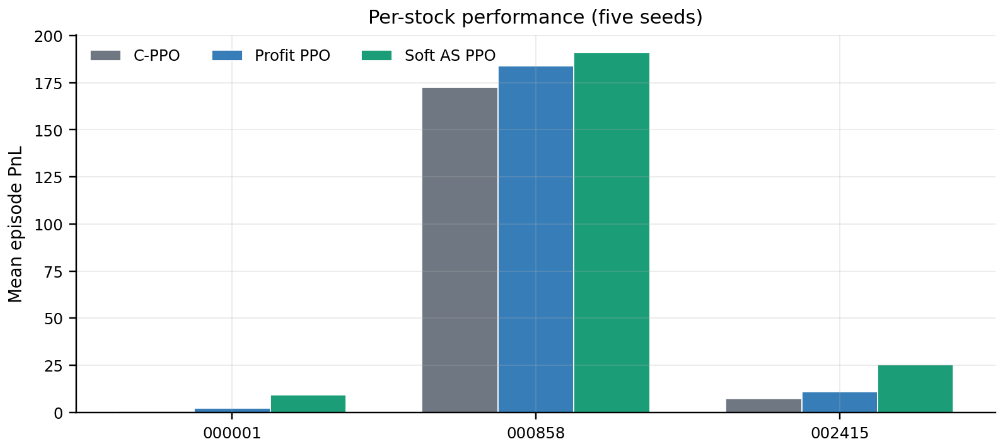

Figure 2: Per-stock mean episode PnL over five seeds. The proposed method improves clearly on the low-price and mid-price stocks and remains competitive on the high-price liquid stock.

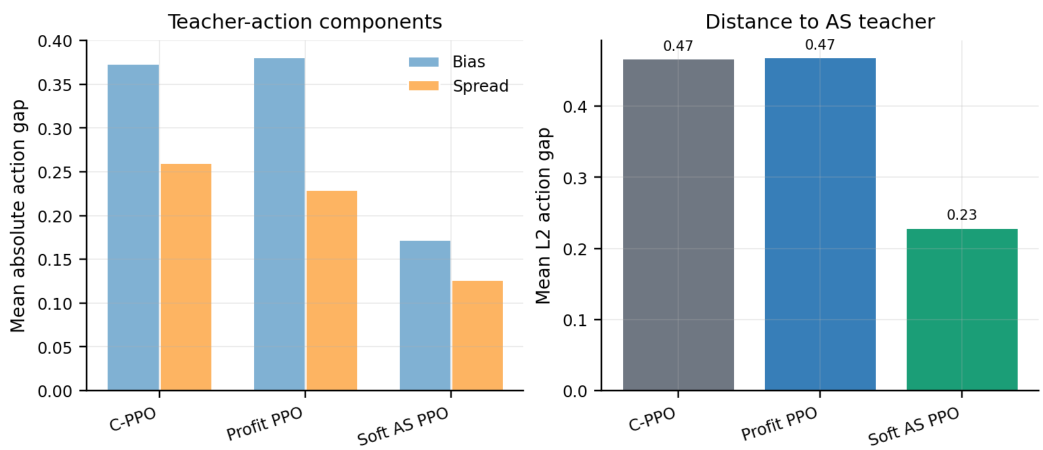

Figure 3: Teacher-divergence diagnostics from saved seed-0/1/2 checkpoints. SOFT-AS PPO remains much closer to the fitted AS teacher than either control, showing that the regularizer directly steers quoting behaviour rather than only changing terminal PnL.

is a useful stress test for the paper: it supports the low-latency result and exposes where an AS-regularized market maker needs stronger inventory or latency-aware controls.

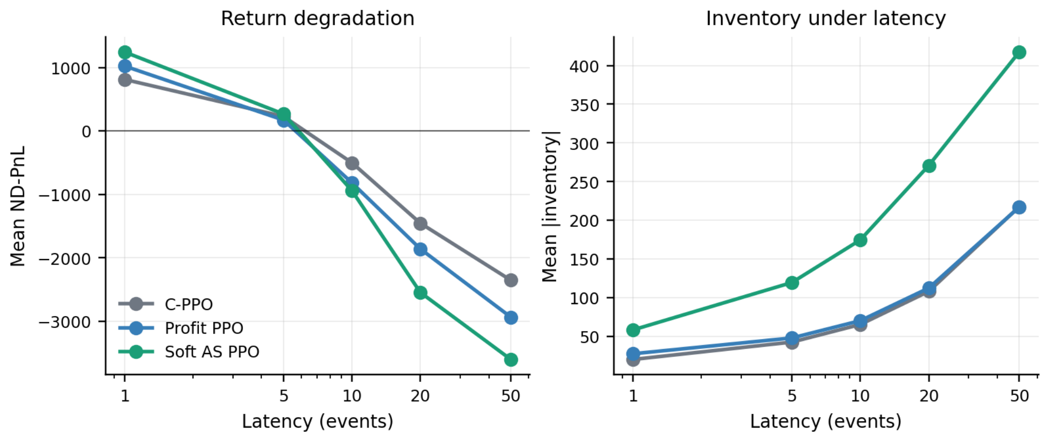

Figure 4: Latency sweep on saved seed-0/1/2 policies. SOFT-AS PPO is best at short latency but is more exposed at high latency because it participates more and carries more inventory.

5.4 Decision trace

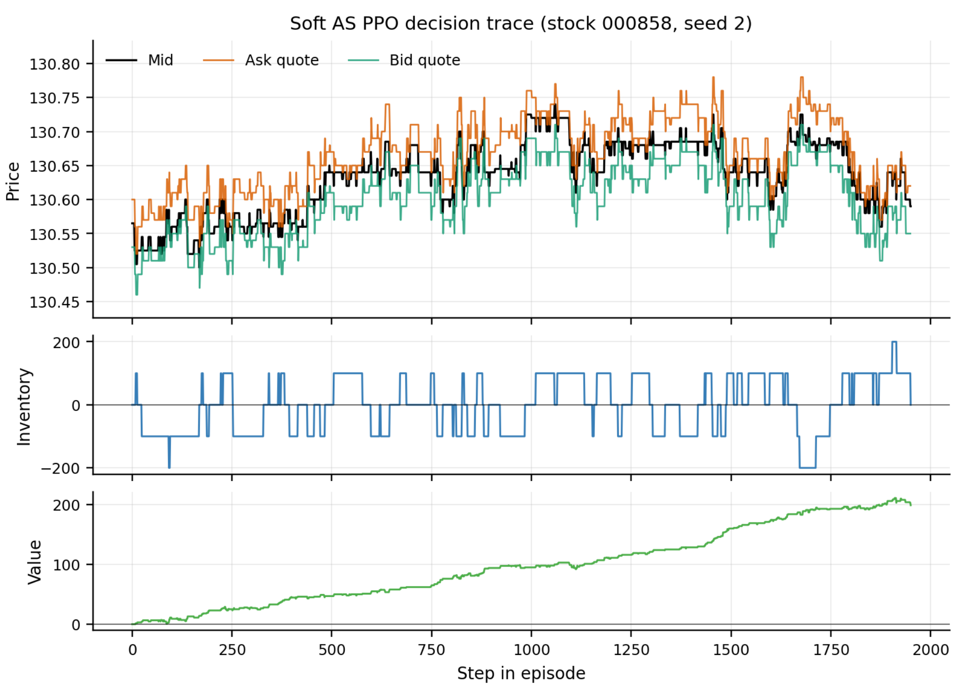

Figure 5: Decision trace for SOFT-AS PPO on stock 000858, seed 2. The policy quotes actively around the mid-price, manages inventory around zero, and accumulates positive marked-to-market value over the episode.

Figure 5 mirrors the original paper's decision-trace diagnostic for the proposed method. It is descriptive rather than causal, but it shows that the improved policy still looks like a market maker rather than a sparse opportunistic trader.

6 Discussion

Why the soft prior helps. The three main methods separate the effect of the old reward from the effect of the AS prior. PROFIT PPO beats C-PPO, so the hybrid reward was a poor fit for this synthetic setting. SOFT-AS PPO then beats PROFIT PPO, which points to the classical teacher adding useful structure beyond the reward change. Hard AS constraints, reported in Appendix B, were weaker because clipping the policy near AS removes profitable deviations. The soft prior works better because the policy can still move away from the teacher when the data support it.

Scope and risk. The claim is intentionally narrow. On a paper-shaped synthetic benchmark, with per-stock policies and matched seeds, AS-regularized PPO is a stronger and more interpretable alternative to hybrid reward shaping. The method trades more and carries more inventory, and the latency sweep shows that this matters when quotes become stale. A live or proprietary-data claim would require a real event stream, transaction costs, market impact, and stricter tail-risk constraints.

Practical value. Reward shaping hides design decisions inside scalar coefficients. AS regularization makes those decisions easier to inspect: the teacher can be calibrated, logged, and stress-tested, and the policy's distance from the teacher can be measured directly. When the teacher is too aggressive, γ, κ, or λ provide meaningful knobs. That makes the approach easier to reason about than a hybrid reward whose terms interact indirectly.

7 Conclusion

We revisited a reward-shaped deep market maker and replaced its hybrid reward with a profit objective plus a soft AS action prior. On a controlled synthetic replication of the original paper's experimental setup, the selected SOFT-AS PPO policy beats both reward-shaped C-PPO and a profit-only PPO control over five seeds per stock. The diagnostics also sharpen the limitation: the method is more active, more inventory-bearing, and more fragile under high latency. Taken together, the experiments make a modest case for AS regularization as a cleaner and more interpretable way to steer deep market-making behaviour than hand-crafted hybrid reward shaping.

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A Synthetic benchmark

The synthetic generator produces event-level LOB days with a latent fair-value process, bid–ask bounce, stable/trending/volatile regimes, market-order flow, and top-10 price/volume levels. Each day has $10{,}000$ raw events before stable-window filtering; downstream evaluation keeps the original paper's stable trading windows, 10:00–11:30 and 13:00–14:30. The three instruments use base prices 16.45 (000001), 130.00 (000858), and 35.00 (002415), giving low-, high-, and mid-price test cases with different realized spread and volatility scales.

The generator first simulates a latent fair value on the tick grid. Regime labels change the probability and size of fair-value moves, and order-flow intensity changes with the regime. Around the fair value, the code builds ask and bid ladders with ten levels of prices and volumes, then records trades whose signs and prices can cross the agent's later quotes during replay. This gives the policy the same kind of 50×40 LOB tensor used by Guo et al. [6], while keeping the data fully reproducible and shareable. The replay simulator has no market impact and no transaction cost, so the benchmark should be read as controlled method-comparison evidence rather than market-deployment evidence.

B Teacher calibration and sensitivity

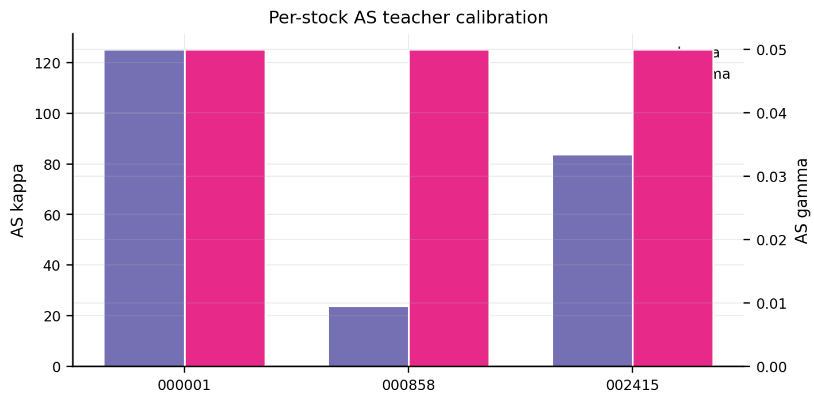


Figure 6: Per-stock AS teacher parameters used by the fitted direct-AS teacher and the AS-regularized policies.

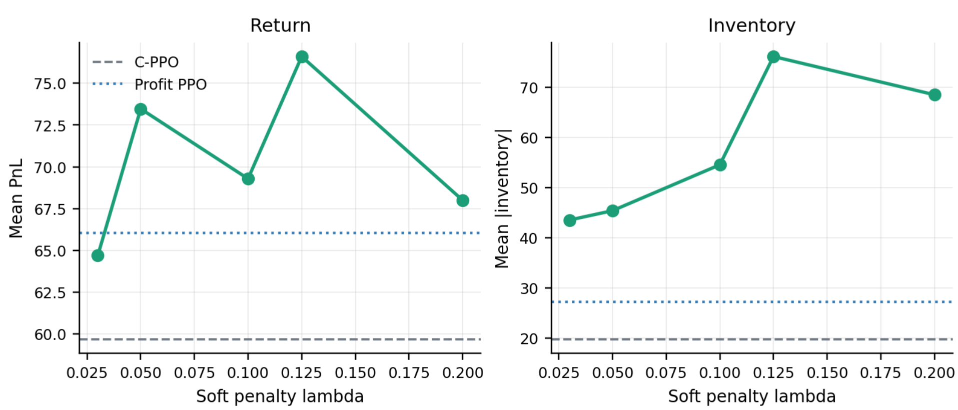


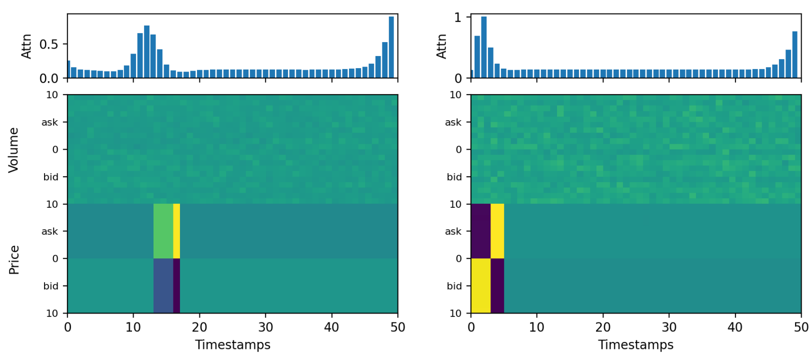
Figure 7: Sensitivity of soft AS regularization to λ in the original three-seed 400k sweep. The selected low-risk teacher is reported in the main paper; this plot shows that the base-teacher penalty weight has a broad useful region, while inventory generally rises with tighter quoting.

The broader matched sweep included base-teacher $\lambda \in \{0.03, 0.05, 0.10, 0.125, 0.20\}$, low-risk and high-risk teachers, fill-κ calibration, hard AS windows, direct AS, and BC warm start. The selected main method uses the low-risk teacher with $\lambda = 0.10$ because it gave the best balance of PnL, Sharpe, and behavioural alignment in the five-seed batch.

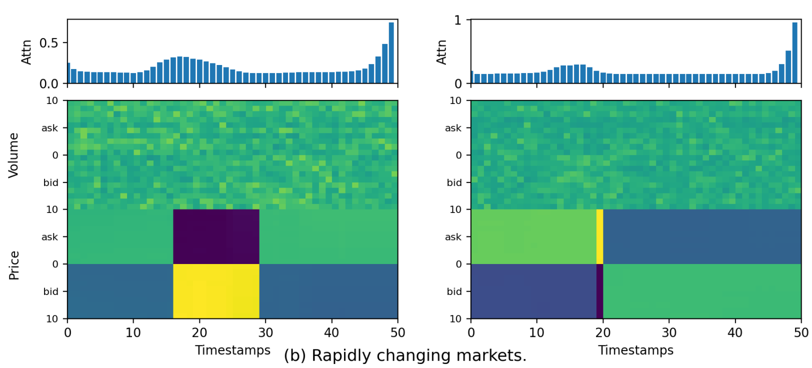
Hard AS applies the teacher as an action constraint rather than a training penalty: the PPO policy proposes an action, and the environment clips that action to a fixed window around the AS action before converting it to quotes. The wider hard window was the best hard-AS variant in the matched batch, but it still underperformed the selected soft regularizer. This is consistent with the role of the constraint. A hard window prevents the policy from taking large deviations even when the synthetic tape makes those deviations profitable. It also treats the AS teacher as locally correct at every step, which is too strong once the fill process, finite action bounds, and end-of-episode liquidation differ from the idealized AS assumptions. Hard AS is therefore useful as a guardrail diagnostic, but it is a less flexible way to combine the teacher with PPO.

BC warm start is different again. In the behavioural-cloning phase, the policy is trained by supervised learning to imitate AS actions on states from the training data; no RL reward is used in that phase. The resulting checkpoint is then fine-tuned with PPO using the pure-profit objective. This tests whether AS is best used only as an initialization. In exploratory runs it helped relative to training from scratch, but it was weaker than keeping the AS prior active during PPO. The likely reason is drift: once PPO fine-tuning begins, the policy can move away from AS without paying any continuing cost.

C Attention diagnostic



(a) Stable markets.



(b) Rapidly changing markets.

Figure 8: Attention diagnostic for the proposed policy. Low-variation and high-variation windows are selected by mid-price standard deviation over the 50-event input window. The plot is included for comparison with the original paper's interpretability figure, but it should be read cautiously: attention maps are descriptive diagnostics, not causal explanations of trading performance.

The windows in Figure 8 are selected mechanically. For every possible 50-event input window, we compute the standard deviation of the level-1 mid-price. The two non-overlapping windows with the lowest values are labelled low-variation, and the two with the highest values are labelled high-variation. This makes the selection reproducible and matches the spirit of the original stable-versus-rapid comparison, but it is still a heuristic. We are confident that the selected windows differ in local mid-price variation under this metric; we should not claim that they are hand-validated economic regimes.

The upper panels show attention mass over the 50 input timestamps, aggregated from the Attn-LOB encoder's self-attention weights. The lower panels show the normalized LOB tensor for the same window. These heatmaps can look unusual because the LOB tensor is not an image: it stacks ask/bid prices and volumes across ten levels, with normalization applied per window. Persistent price ladders and large volume bands therefore appear as blocks or stripes. Attention can also look fairly flat after aggregation because different heads attend to different timestamps, and summing or averaging over heads smooths those patterns. In this run the diagnostic mainly says that the encoder is using broad recent context rather than placing a sharp, paper-like spike on a single event. That is informative, but weaker than the attention story in the original paper and should be treated as a qualitative check rather than evidence of mechanism.

D Per-stock numerical results

Table 3: Per-stock means over five seeds.

Stock	Method	PnL	ND-PnL	Sharpe	Abs. inv.
000001	C-PPO	0.44	8.0	0.19	0.5
000001	PROFIT PPO	2.16	44.3	0.69	2.2
000001	SOFT-AS PPO	9.16	282.7	0.75	64.2
000858	C-PPO	172.64	2292.5	4.66	52.1
000858	PROFIT PPO	183.83	2746.6	4.38	69.2
000858	SOFT-AS PPO	190.92	2689.4	4.67	62.4
002415	C-PPO	7.15	135.6	1.36	5.1
002415	PROFIT PPO	11.02	240.5	1.81	8.3
002415	SOFT-AS PPO	25.44	699.4	2.20	36.2